

Naive Bayes Foundations: Mutually Exclusive Events

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Conditional probability ([Note 82](#)).
- **Leads to:** Naive Bayes ([Note 87](#)); Voting ensembles ([Note 102](#)).

1. Overview

Key point: Two events are mutually exclusive when they cannot happen together: $P(A \cap B) = 0$. Mutual exclusivity is a different idea from independence, and the two are easily confused.

The previous Note defined **independent** events (G-934): they can happen together, but one does not affect the other. **Mutually exclusive** events are different: they can never happen together. The two ideas are easy to mix up, so this Note puts them side by side.

2. The definition

Key point: Mutually exclusive: $P(A \cap B) = 0$, so $P(A | B) = 0$.

Two events A and B are **mutually exclusive** (G-1286) if

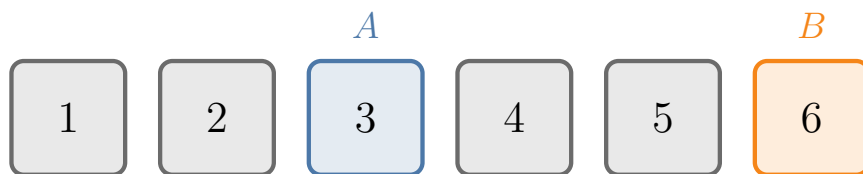
$$P(A \cap B) = 0$$

They have no outcome in common, so at most one of them can happen.

Examples:

- A single coin toss cannot land both heads and tails.
- At a junction, a driver turns either left or right, not both.
- One die cannot show both 3 and 6 on the same roll.

one roll of one die



no face is in both: $P(A \cap B) = 0$

given B , only face 6 is left: $P(A \mid B) = 0$

A or B : 2 of 6 faces, $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

Figure 1: One roll of one die, with A = “shows 3” and B = “shows 6”. The two events share no face

Figure 1 shows the die example: A and B sit on different faces, so no outcome is in both. Once we know B happened, the roll is face 6, and face 3 is ruled out.

3. Conditional probability for mutually exclusive events

Key point: If B happened, A cannot have happened: $P(A \mid B) = 0$.

Putting $P(A \cap B) = 0$ into the conditional probability formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0$$

With numbers: die 1 has shown 3. The probability that the same die shows 6 on that roll is 0. A driver has turned left; the probability that they turned right is 0.

Figure 2 checks this on a million simulated rolls. Over all rolls, a 3 comes up 0.1667 of the time. Among the rolls that showed 6, not one showed 3: the share is exactly 0. The right panel previews the addition rule of the Extra in Section 4.

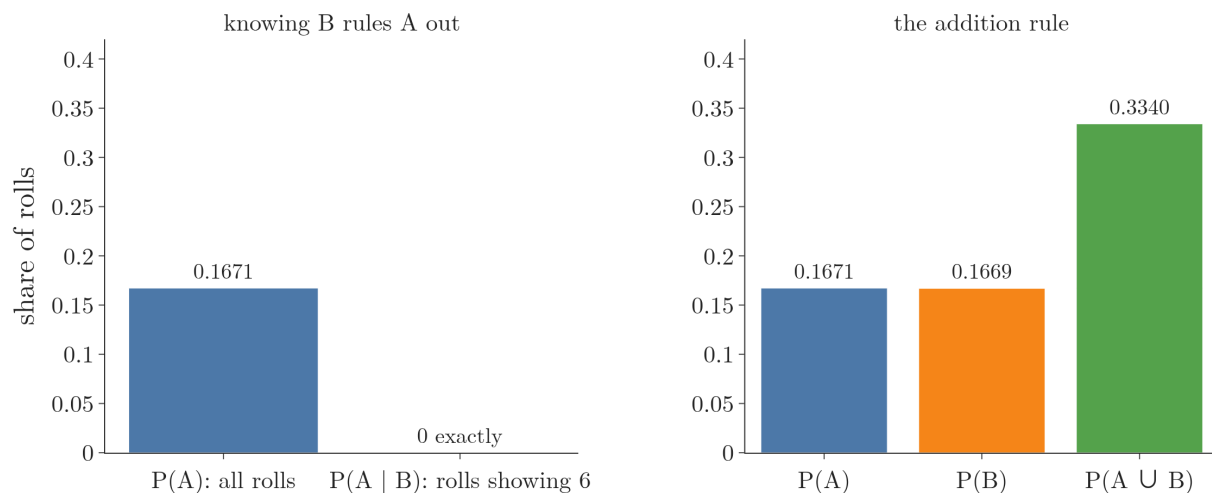


Figure 2: One million rolls of a fair die. Left: the share showing 3 is 0.1667 over all rolls, and exactly 0 among the rolls that showed 6. Right: the shares of 3, of 6, and of “3 or 6”, which is their sum, about $1/3$.

4. Mutually exclusive is not independent

Key point: Knowing B happened tells us A did not, which is a lot of information. So mutually exclusive events (each with a positive probability) are always dependent.

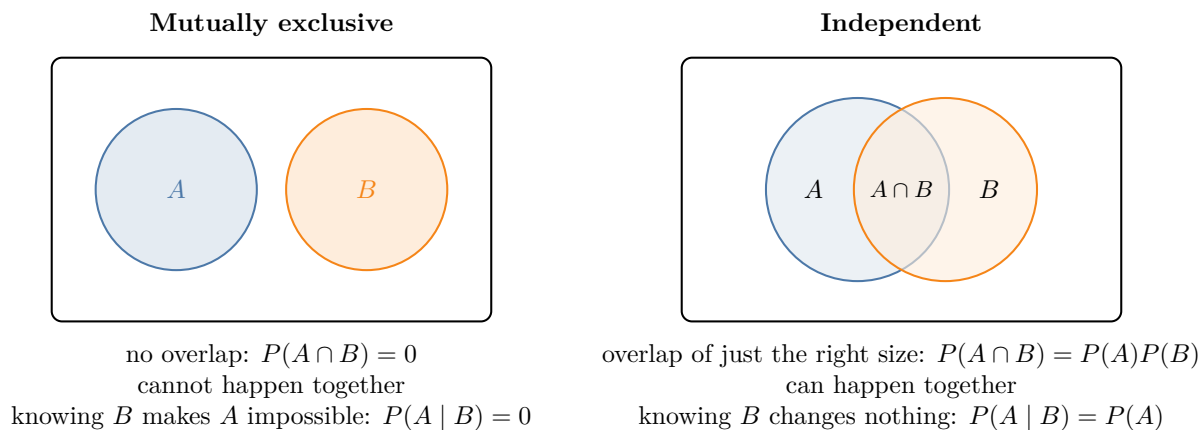


Figure 3: Mutually exclusive versus independent events

Figure 3 lays the two ideas side by side.

	Mutually exclusive	Independent
Can happen together?	no	yes
$P(A \cap B)$	0	$P(A) \times P(B)$
$P(A B)$	0	$P(A)$
Does B tell us about A?	yes: A did not happen	no

For “die 1 shows 3” (A) and “die 1 shows 6” (B), each has probability $1/6$:

- $P(A \cap B) = 0$, so they are mutually exclusive.
- $P(A) \times P(B) = 1/36 \neq 0$, so they are **not** independent.

In fact, if both events have a probability above 0, being mutually exclusive always rules out being independent. Learning that one happened tells us for certain that the other did not.

An everyday picture: a light switch is either on or off. The two states are mutually exclusive, and for that very reason, hearing “the switch is on” tells us everything about “off”.

Extra: For mutually exclusive events, the probability that one **or** the other happens, their **union** (G-2045), is a simple sum: $P(A \cup B) = P(A) + P(B)$. Here $1/6 + 1/6 = 1/3$. This **addition rule** (G-173) is used in the Bayes’ theorem Notes to split a probability into separate cases.

5. Summary

- Mutually exclusive: $P(A \cap B) = 0$; the events cannot happen together, and $P(A | B) = 0$.
- Independent: $P(A \cap B) = P(A)P(B)$; they can happen together, and $P(A | B) = P(A)$.
- Mutually exclusive events with positive probabilities are never independent.

6. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 3 | Mutually Exclusive Events”, YouTube, <https://www.youtube.com/watch?v=nneTjTYikBE>

7. Key terms

Term	Meaning
Mutually exclusive events	Events that cannot happen at the same time; their intersection has probability 0
Union ($A \cup B$)	The event that A or B (or both) happens
Addition rule	For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$